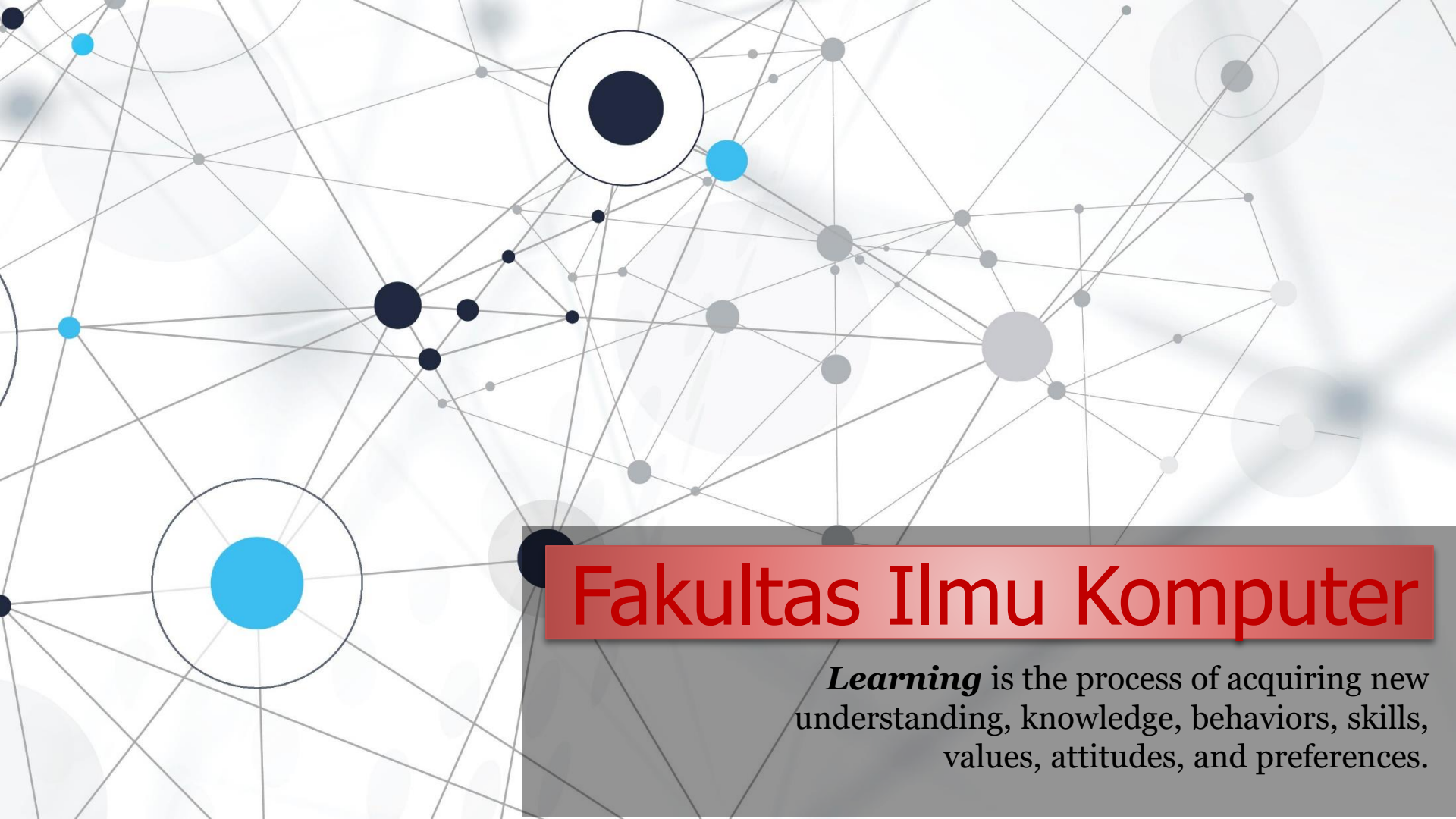


An abstract network diagram with various sized nodes (black, blue, and grey) connected by thin grey lines. Some nodes are highlighted with larger circles. The background is white with faint grey circles.

Fakultas Ilmu Komputer

Learning is the process of acquiring new understanding, knowledge, behaviors, skills, values, attitudes, and preferences.

PART II # Continuous Integration (CI)



Welcome to *the* DevOps Engineering *course*

Chapter #1
DevOps Culture *and* Practices

1 - Continuous Integration (CI)



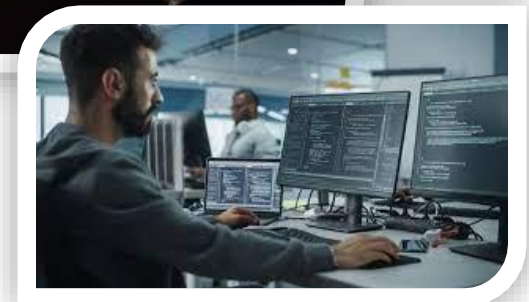
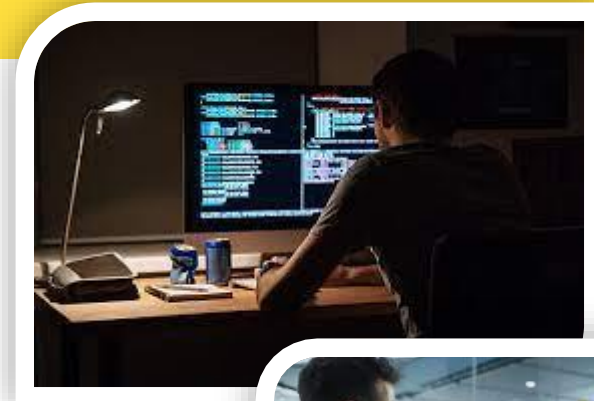
Software Engineer

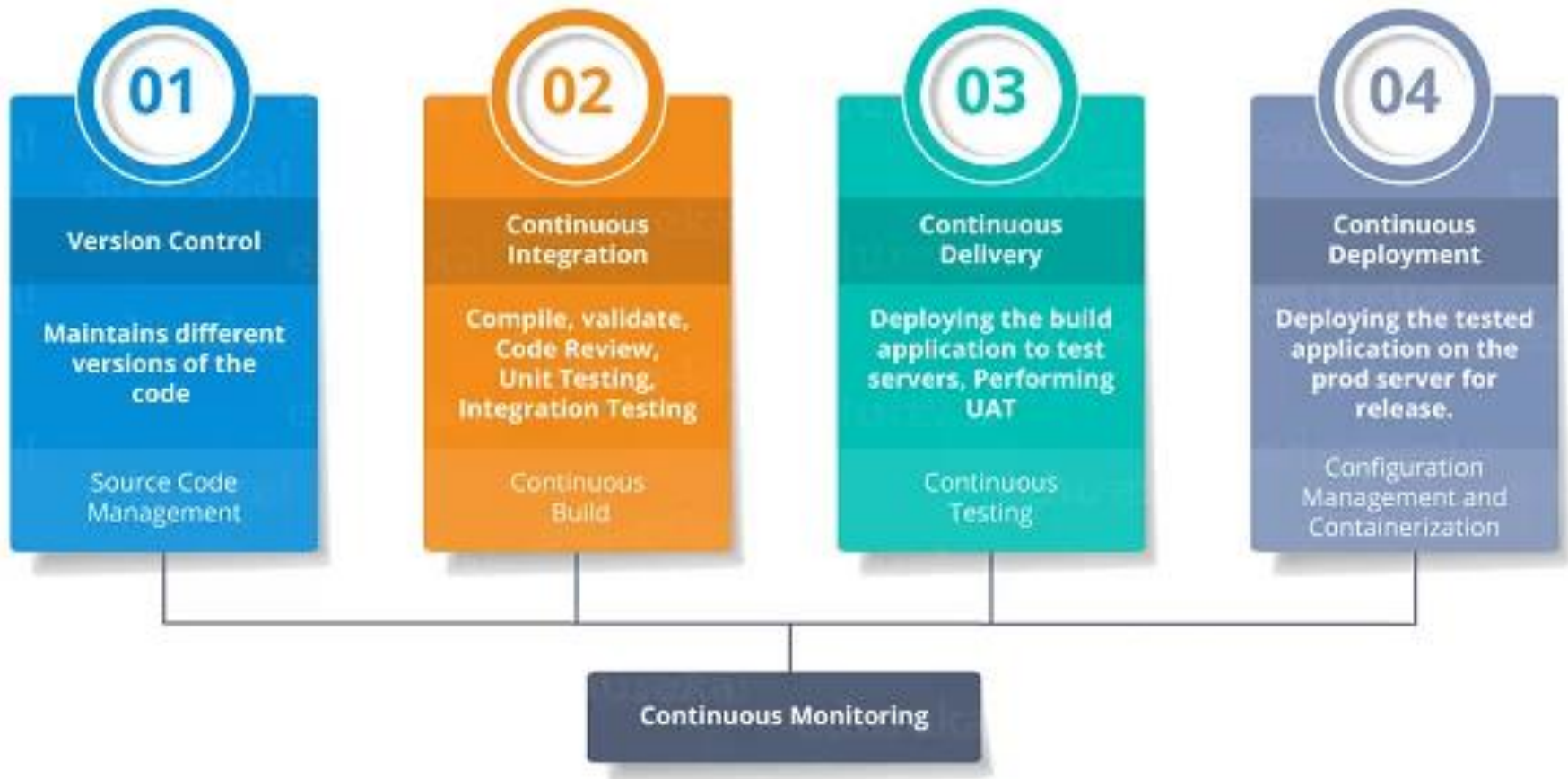
Orang yang mendesain, membangun, memelihara, menguji, dan mengevaluasi perangkat lunak komputer menggunakan konsep rekayasa perangkat lunak.

VS

Software Developer

Orang yang bertugas merancang, membuat, dan meluncurkan program komputer, bisa berupa operation system software atau application software





In software development, DevOps is one approach that integrates **development**, **testing**, **integration**, and **monitoring** in a continuous or real-time manner. A software developer certainly wants the software created to be more accessible to customers, don't you think? Therefore, the CI/CD Pipeline acts as a tool **to drive the software development process** and **reduce risks at each stage of development**, making it easier to achieve the ultimate goal of reaching customers.

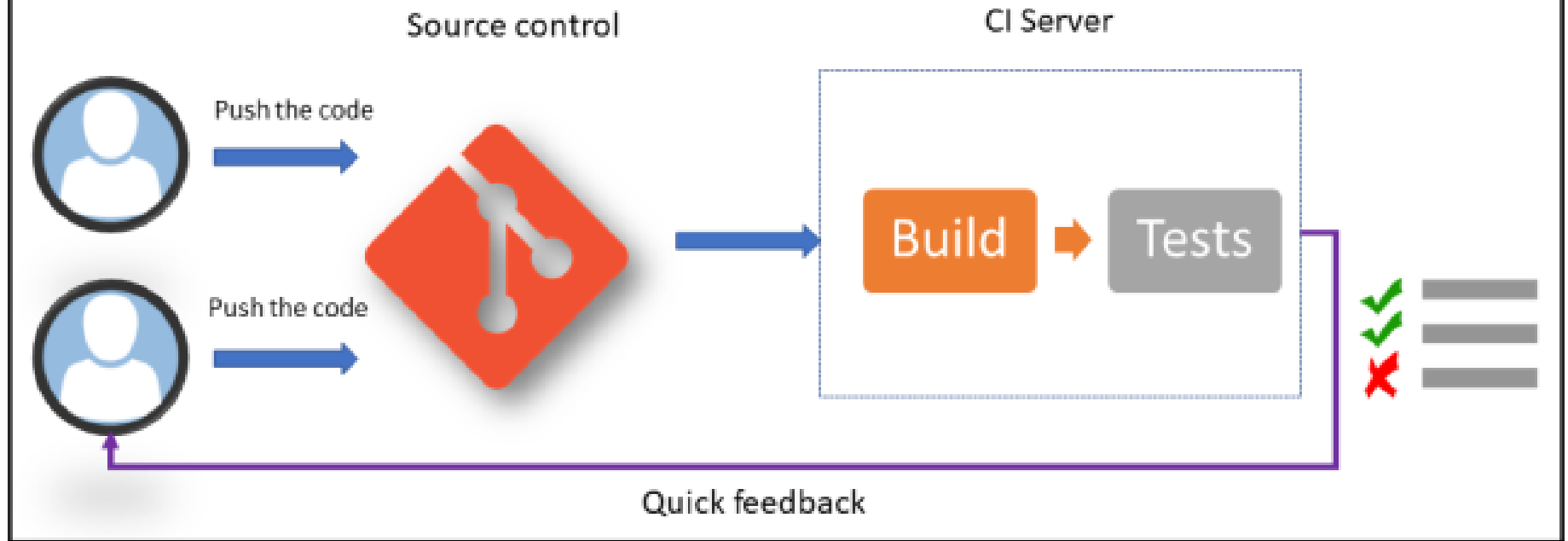
Continuous Integration (CI)

In the following definition given by **Martin Fowler**, there are three key things mentioned, members of a team, integrate, and as quickly as possible:

"Continuous Integration is a software development practice where members of a team integrate their work frequently... Each integration is verified by an automated build (including test) to detect integration errors as quickly as possible."

- ❖ With the spirit of collaboration and communication, because the execution of CI impacts all members in terms of work methodology and therefore collaboration.
- ❖ CI requires the implementation of processes (*branch, commit, pull request, code review, and so on*) with automation that is done with tools adapted to the whole team (**Git**, **Jenkins**, team city, gitlab, circle CI, Azure DevOps, and so on).
- ❖ Finally, CI must run quickly to collect feedback on code integration as soon as possible and hence be able to deliver new features more quickly to users.

The continuous integration (CI) pipeline

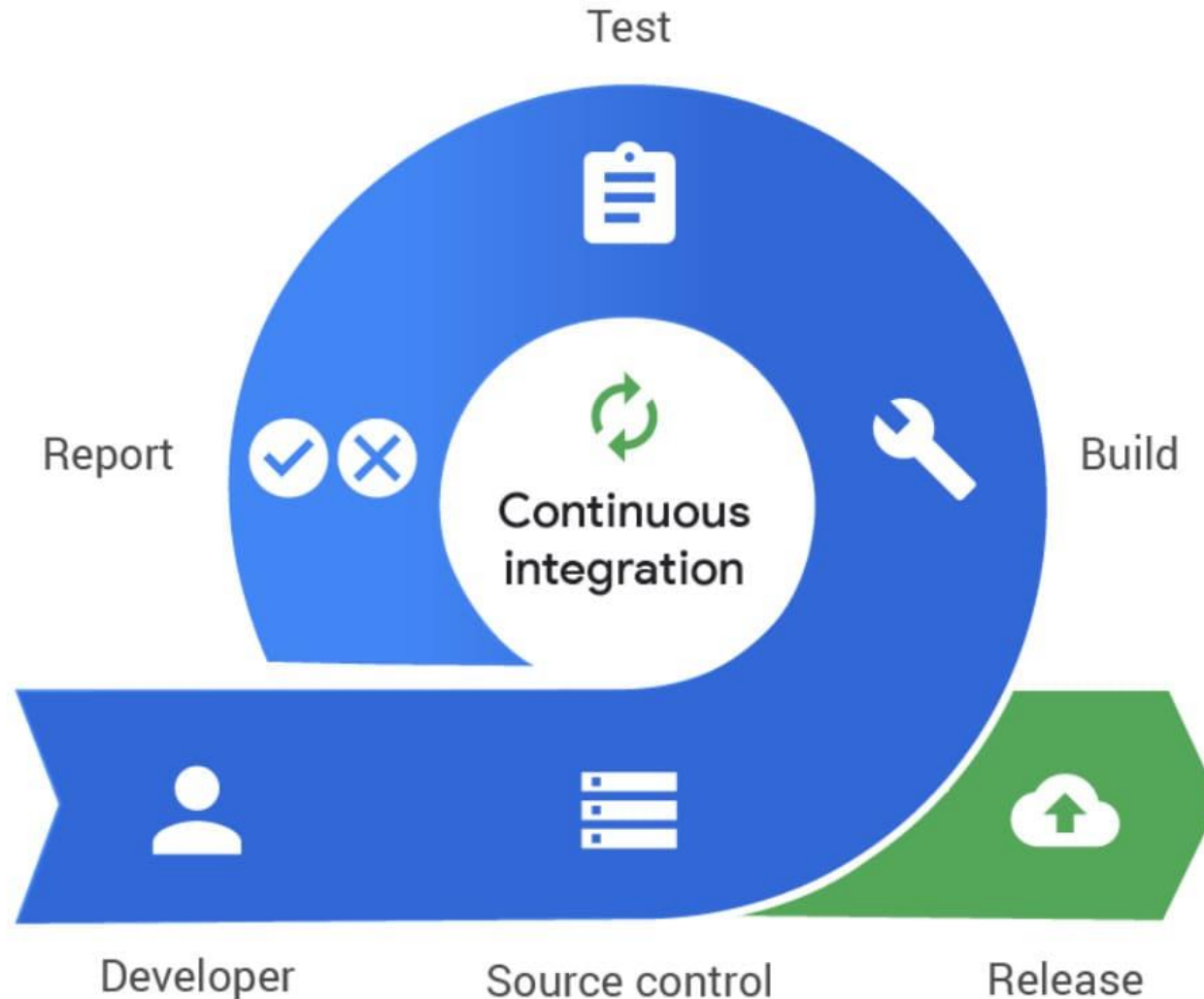


- ❖ This **CI process must be optimized as soon as possible** so that it can run fast and developers can have quick feedback on the integration of their code.
- ❖ For example, code that is archived and does not compile or whose **test execution fails can impact and block the entire team.**
- ❖ This diagram shows the cyclical steps of continuous integration that include the code being pushed into the **Source Code Manager (SCM)** by the *team members* and the execution of the build and test by the **CI server**, and the purpose of this fast process is to provide rapid feedback to members.

Bad Practices == Failure Tests

- 1) Bad practices of CI can result in the failure of tests in the CI, deactivating the test execution, taking as arguments: it is not serious, it is necessary to deliver quickly, or the code that compiles it is essential.
- 2) This practice can have serious consequences when the errors detected by the tests are revealed in production.
- 3) The time saved during CI will be lost on fixing errors with hotfixes and redeploying them quickly with stress.

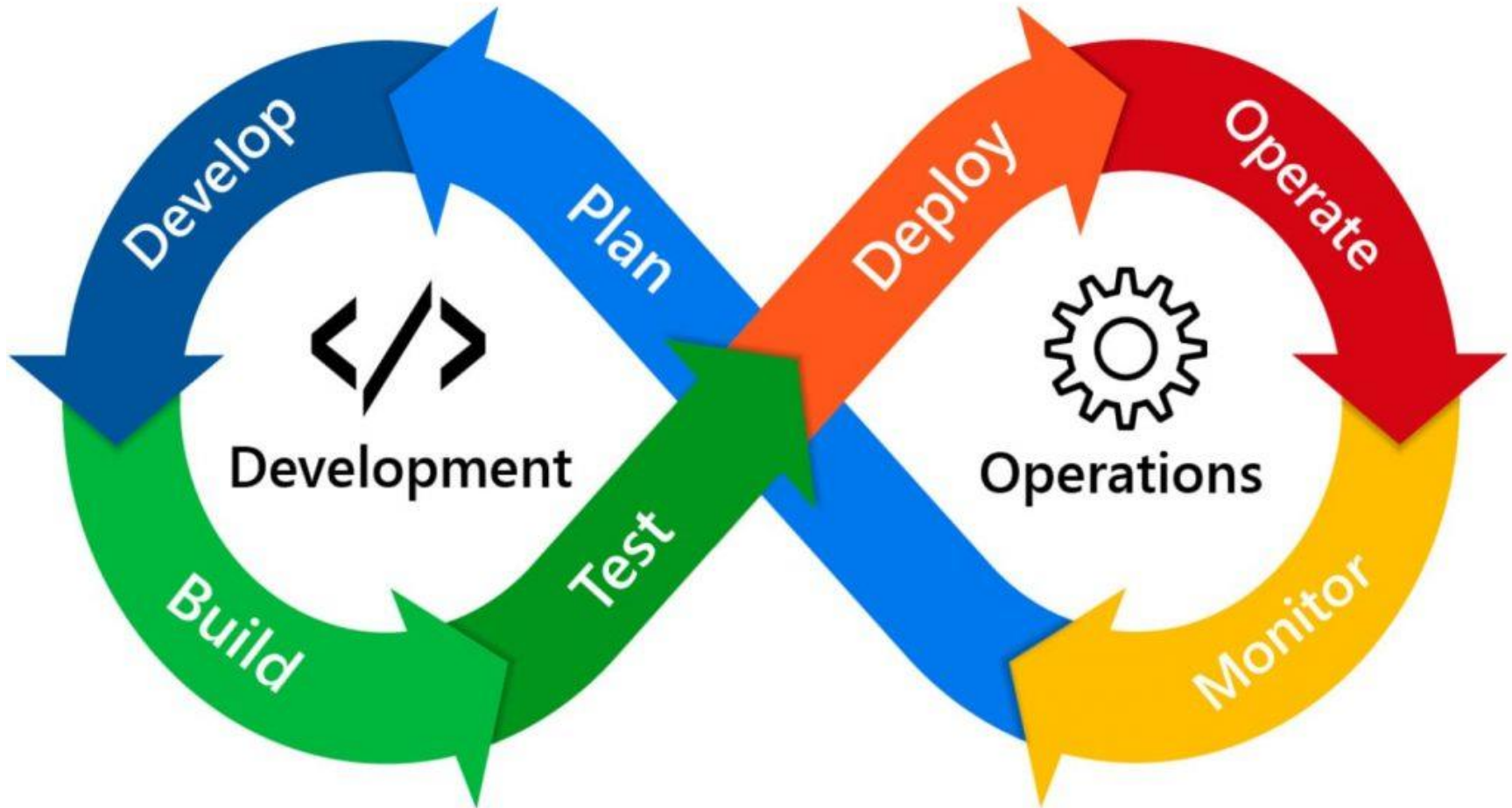
This process is the opposite of DevOps culture with ***poor application quality*** for end users and no real feedback, and, instead of developing new features, **we spend time correcting errors.**



Continuous Integration (CI) aims to make it easier for developers to integrate their work continuously. Especially in this day and age, many companies employ many developers at once for **time efficiency**.

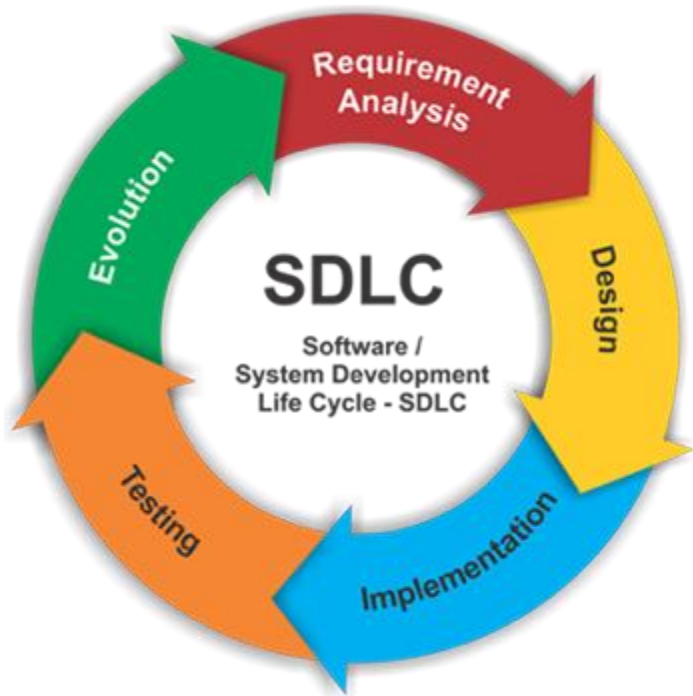
The slide features a white background with green leaves and branches framing the top and bottom edges. The leaves are vibrant green and appear to be from a tree or shrub. The text is centered in the middle of the slide.

2 - Automation Server *in* DevOps



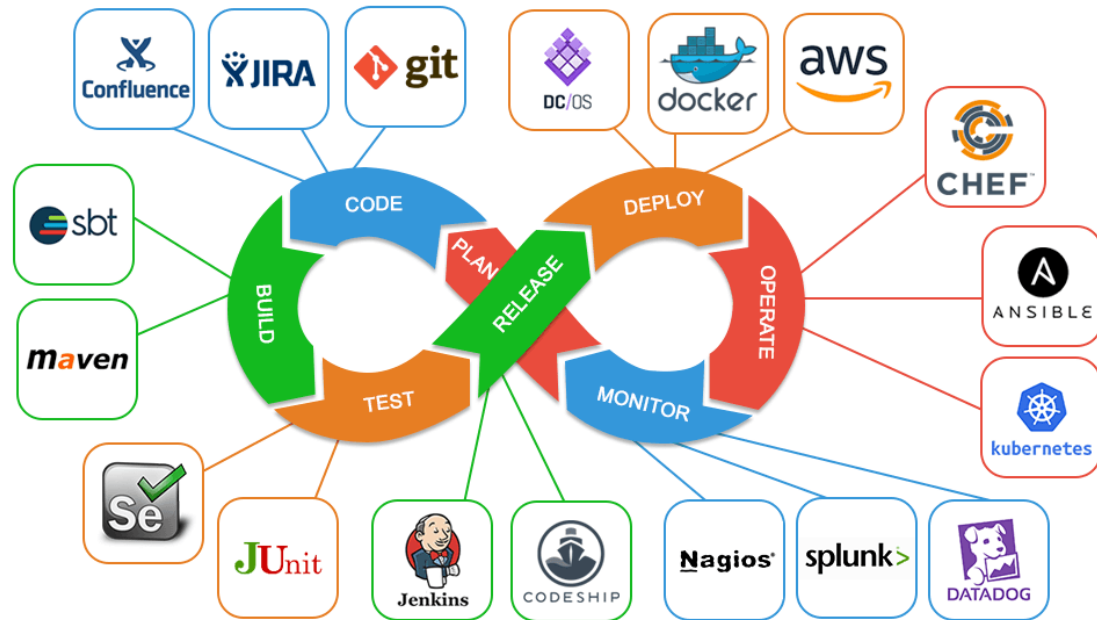
CI/CD adalah singkatan dari Continuous Integration, Continuous Delivery, dan Continuous Deployment. **CI/CD** adalah upaya mengubah **proses pengembangan software yang tadinya manual menjadi otomatis**. CI/CD adalah metode pengembangan software yang memudahkan, mempercepat, sekaligus meningkatkan kualitas hasil akhir software yang dirilis.

(Before) Automation Servers



- 1) Dalam pengembangan software (*perangkat lunak*), koding hanyalah tahapan awal saja.
- 2) Setelah koding, masih banyak hal yang biasanya dilakukan oleh *software developer* (*perangkat lunak*) sebelum aplikasi yang dibuat sampai ke *production server*.
- 3) Sebelum adanya Automation Server, biasanya semuanya dilakukan **secara manual**, namun resikonya adalah sering terjadi **kesalahan dan tidak konsisten**.

Contoh menjalankan ***unit test***, download dependency aplikasi, setup database, membuat distribusi aplikasi, dan lain-lain. Hal-hal diatas sangat kompleks jika harus dilakukan secara manual terus menerus.

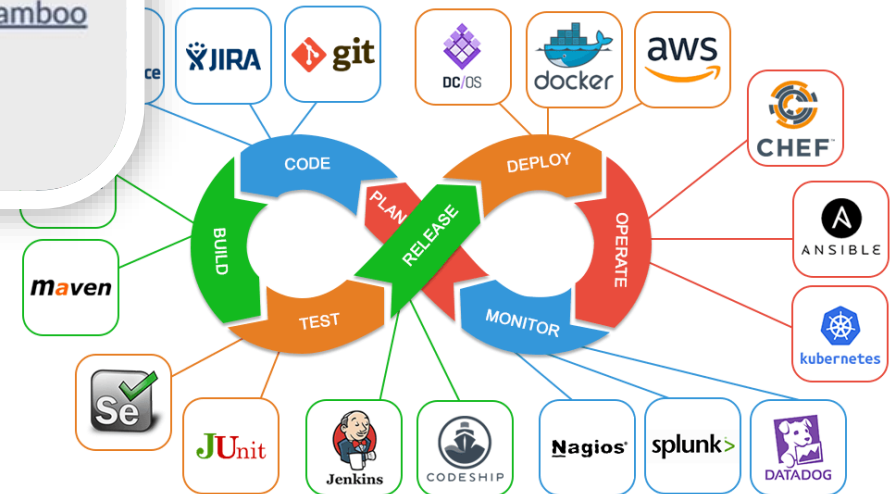


- 1) **Automation Server** adalah tool/aplikasi yang bisa kita gunakan untuk **melakukan proses otomatisasi perintah** yang kita instruksikan.
- 2) Ada banyak sekali keuntungan yang kita dapatkan jika menggunakan Automation Server, dibanding manual dalam tahapan pengembangan perangkat lunak.
- 3) (a) Automation Server **mempercepat proses**, karena kita tidak perlu menunggu di Laptop kita.
(b) Automation Server bisa **berjalan konsisten**, di Laptop programmer mungkin perangkat lunak bisa berbeda-beda.
- 4) Automation Server bisa **berjalan secara otomatis** (hampir sama seperti *scheduler*), bisa memberitahu kita ketika terjadi kesalahan, dan lain-lain.

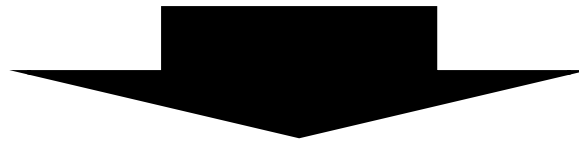
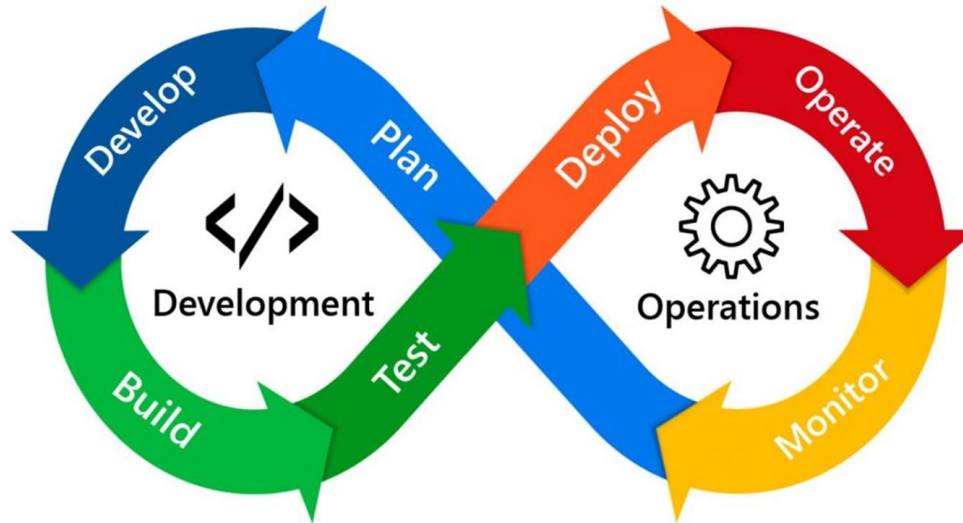
Automation Tools DevOps 2023

Contoh Automation Server

- Jenkins <https://www.jenkins.io/>
- Team City <https://www.jetbrains.com/teamcity/>
- GitLab CI/CD <https://docs.gitlab.com/ee/ci/>
- Github Action <https://github.com/features/actions>
- Atlassian Bamboo <https://www.atlassian.com/software/bamboo>
- Circle CI <https://circleci.com/>
- Travis CI <https://travis-ci.org/>
- Dan lain-lain



Jenkins adalah tools CI/CD yang mendukung **otomatisasi pengembangan software** dari mulai code merging hingga *deployment*. Selain itu, Jenkins juga menyediakan ratusan plugin tambahan, serta mendukung berbagai jenis OS server, seperti **Windows, Linux, dan MacOS**.



+



Jenkins

What is Jenkins?

In DevOps, an **automation server** refers to a software tool or platform that plays a crucial role in automating various aspects of the software development and deployment process. Automation servers are often used to streamline and manage tasks such as continuous integration (CI) and continuous delivery (CD), which are essential practices in DevOps.

- 1) **Jenkins** adalah **Open Source Automation Server** yang paling populer saat ini (2023).
- 2) Sebelum Jenkins, awalnya *project* ini bernama **Hudson**.
- 3) Hudson dibuat menggunakan Java, dan pengembang inti dari Hudson kebanyakan bekerja di Sun Microsystem (pemiliki Java).
- 4) Tahun 2011, terjadi sengketa setelah pembelian Sun Microsystem oleh Oracle, dan karena trademark Hudson dipegang oleh Oracle, akhirnya pengembang inti dari Hudson membuat project baru bernama Jenkins dari **source code Hudson**.
- 5) Hudson sendiri akhirnya didonasikan ke Eclipse Foundation tahun 2012 dan akhirnya pengembangannya berhenti tahun 2017.
- 6) Jenkins official website: <https://www.jenkins.io/>



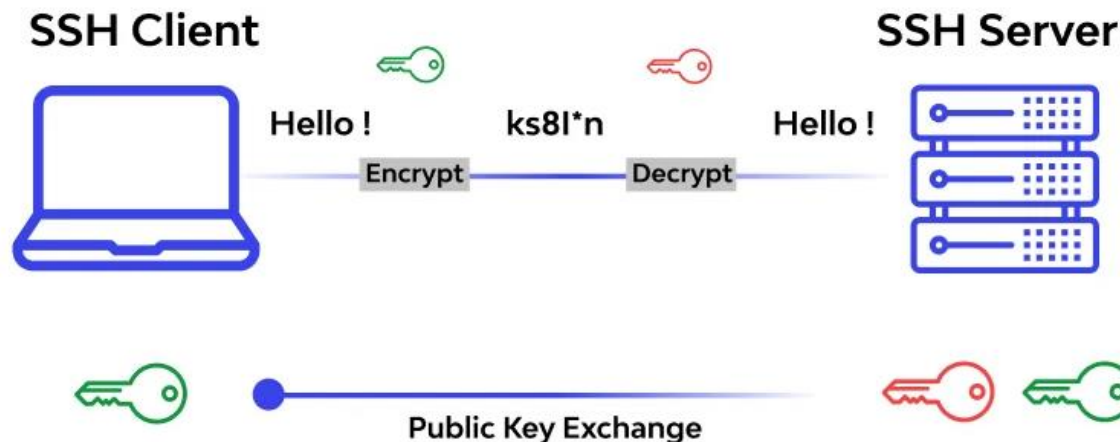
Software & Hardware Requirements

[1] Hardware Requirements for Jenkins (Recommended):

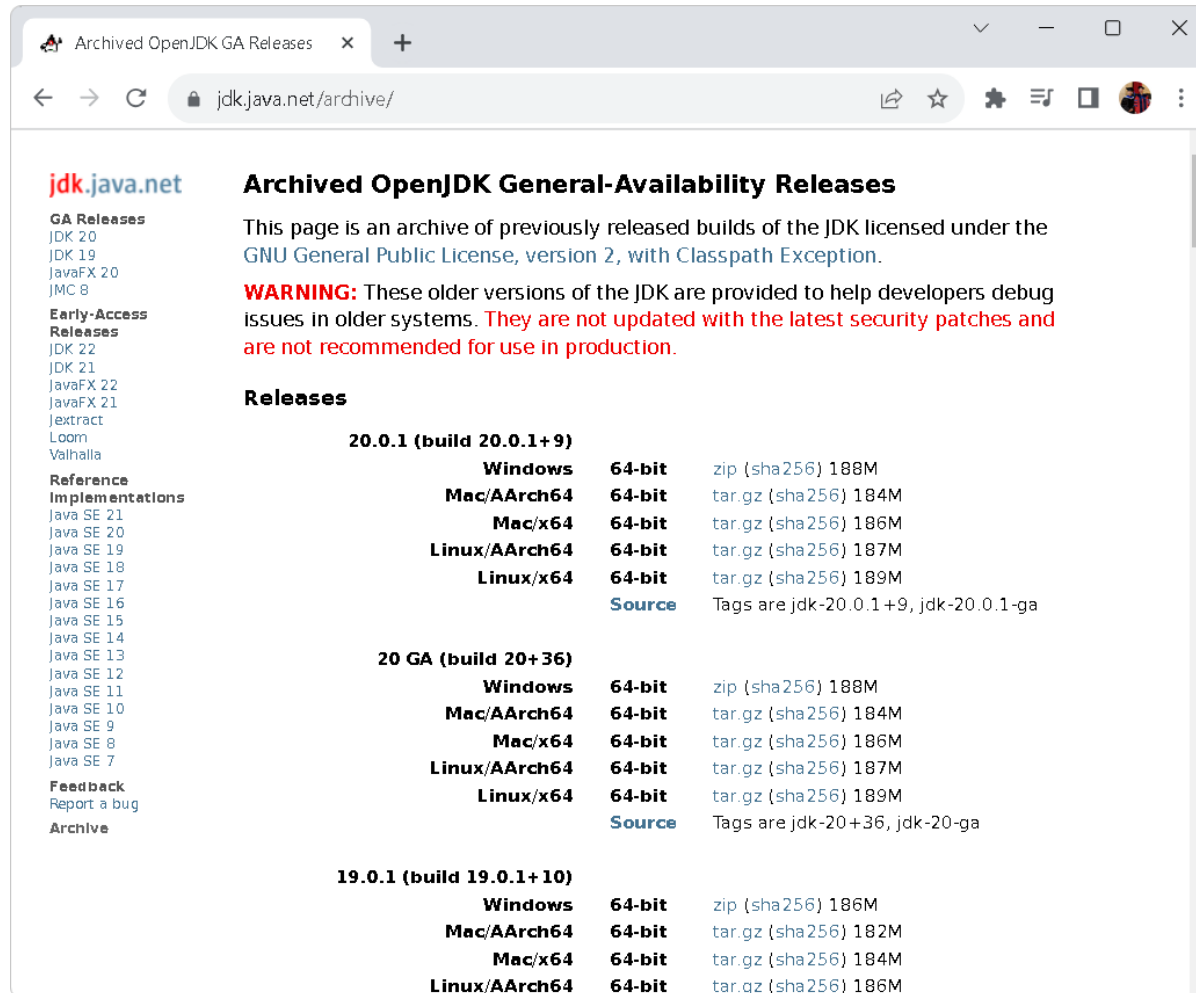
- ❖ 4 GB+ of RAM.
- ❖ 50 GB+ of Drive Space.

[2] Software Requirements for Jenkins (Recommended):

- ❖ Secure Shell Protocol (SSH) => Cloud Server Linux.
- ❖ Java versi 11 => <http://jdk.java.net/archive/>
- ❖ Git + Github => <https://git-scm.com/>
- ❖ Web Browser.



Java Version 11 (Recommended)



The screenshot shows a web browser window with the address bar displaying `jdk.java.net/archive/`. The page title is "Archived OpenJDK GA Releases". The left sidebar contains a navigation menu with links to "GA Releases", "Early-Access Releases", and "Reference Implementations". The main content area is titled "Archived OpenJDK General-Availability Releases" and contains a warning about older versions. Below the warning, there are three sections of releases: "20.0.1 (build 20.0.1+9)", "20 GA (build 20+36)", and "19.0.1 (build 19.0.1+10)". Each section lists download links for Windows, Mac/AArch64, Mac/x64, Linux/AArch64, and Linux/x64, along with the file format, architecture, and size.

jdk.java.net

- GA Releases
 - JDK 20
 - JDK 19
 - JavaFX 20
 - JMC 8
- Early-Access Releases
 - JDK 22
 - JDK 21
 - JavaFX 22
 - JavaFX 21
 - Jextract
 - Loom
 - Valhalla
- Reference Implementations
 - Java SE 21
 - Java SE 20
 - Java SE 19
 - Java SE 18
 - Java SE 17
 - Java SE 16
 - Java SE 15
 - Java SE 14
 - Java SE 13
 - Java SE 12
 - Java SE 11
 - Java SE 10
 - Java SE 9
 - Java SE 8
 - Java SE 7
- Feedback
 - Report a bug
- Archive

Archived OpenJDK General-Availability Releases

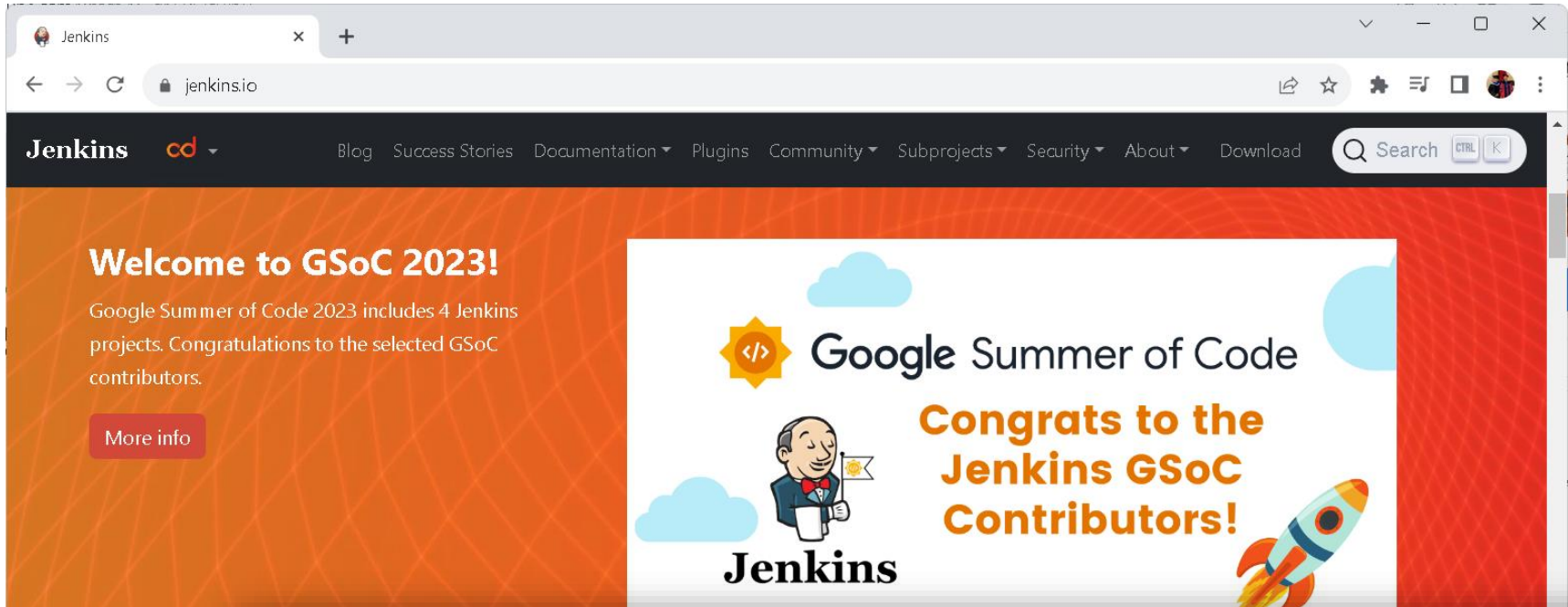
This page is an archive of previously released builds of the JDK licensed under the GNU General Public License, version 2, with Classpath Exception.

WARNING: These older versions of the JDK are provided to help developers debug issues in older systems. **They are not updated with the latest security patches and are not recommended for use in production.**

Releases

20.0.1 (build 20.0.1+9)			
Windows	64-bit	zip (sha256)	188M
Mac/AArch64	64-bit	tar.gz (sha256)	184M
Mac/x64	64-bit	tar.gz (sha256)	186M
Linux/AArch64	64-bit	tar.gz (sha256)	187M
Linux/x64	64-bit	tar.gz (sha256)	189M
	Source	Tags are jdk-20.0.1+9, jdk-20.0.1-ga	
20 GA (build 20+36)			
Windows	64-bit	zip (sha256)	188M
Mac/AArch64	64-bit	tar.gz (sha256)	184M
Mac/x64	64-bit	tar.gz (sha256)	186M
Linux/AArch64	64-bit	tar.gz (sha256)	187M
Linux/x64	64-bit	tar.gz (sha256)	189M
	Source	Tags are jdk-20+36, jdk-20-ga	
19.0.1 (build 19.0.1+10)			
Windows	64-bit	zip (sha256)	186M
Mac/AArch64	64-bit	tar.gz (sha256)	182M
Mac/x64	64-bit	tar.gz (sha256)	184M
Linux/AArch64	64-bit	tar.gz (sha256)	186M

Jenkins Official Website



Continuous Integration and Continuous Delivery

As an extensible automation server, Jenkins can be used as a simple CI server or turned into the continuous delivery hub for any project.



Easy installation

Jenkins is a self-contained Java-based program, ready to run out-of-the-box, with packages for Windows, Linux, macOS and other Unix-like operating systems.



Easy configuration

Jenkins can be easily set up and configured via its web interface, which includes on-the-fly error checks and built-in help.



Plugins

With hundreds of plugins in the Update Center, Jenkins integrates with practically every tool in the continuous integration and continuous delivery toolchain.



Extensible

Jenkins can be extended via its plugin architecture, providing nearly infinite possibilities for what Jenkins can do.



Distributed

Jenkins can easily distribute work across multiple machines, helping drive builds, tests and deployments across multiple platforms faster.



Exercise *for* Students

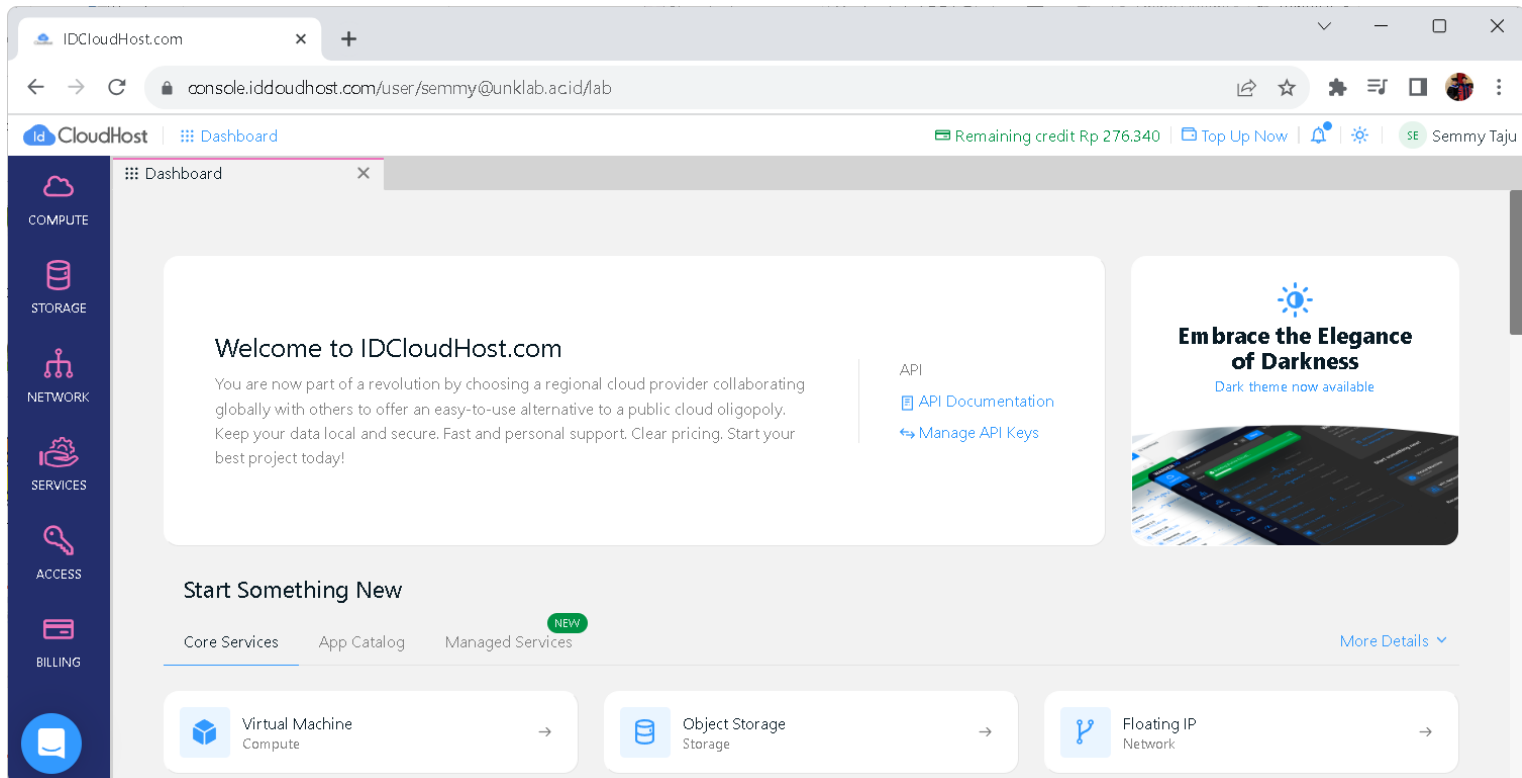
A decorative border of green leaves and branches at the top of the page.

1 – Cloud Server *in* Ubuntu 22.04

A decorative border of green leaves and branches at the bottom of the page.



Start *with* New VPS (Cloud Server)



Hardware Requirements *for Jenkins* (Recommended):

- ❖ 4 GB+ of RAM.
- ❖ 50 GB+ of Drive Space.

[1] Install Apache2 *via* Putty

[1] Change hak akses *superuser* atau *root*:

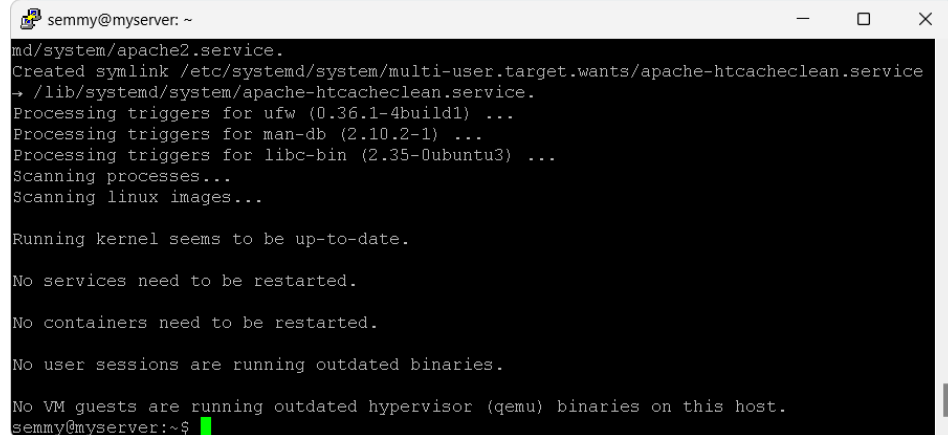
➤ ***sudo su***

[2] Install Apache di linux ubuntu 22.04, dengan menjalankan perintah berikut:

➤ ***sudo apt install apache2***

Keterangan:

- 1) Perintah ***sudo*** adalah singkatan dari "*Superuser Do*". Perintah ini digunakan untuk memberikan akses root atau superuser pada pengguna yang tidak memiliki izin tersebut.
- 2) Perintah ***su***, di sisi lain, adalah singkatan dari "*Substitute User*". Perintah ini digunakan untuk beralih ke akun pengguna lain tanpa harus keluar dari sesi terminal saat ini.
- 3) Perintah ***apt*** adalah singkatan dari "*Advanced Packaging Tool*". Perintah *apt* merupakan sebuah tool pada sistem operasi Debian dan turunannya, seperti Ubuntu, yang digunakan untuk mengelola paket software, yaitu menginstal, menghapus, atau memperbarui paket software pada sistem operasi.



```
semmy@myserver: ~  
md/system/apache2.service.  
Created symlink /etc/systemd/system/multi-user.target.wants/apache-htcacheclean.service  
→ /lib/systemd/system/apache-htcacheclean.service.  
Processing triggers for ufw (0.36.1-4build1) ...  
Processing triggers for man-db (2.10.2-1) ...  
Processing triggers for libc-bin (2.35-0ubuntu3) ...  
Scanning processes...  
Scanning linux images...  
  
Running kernel seems to be up-to-date.  
  
No services need to be restarted.  
  
No containers need to be restarted.  
  
No user sessions are running outdated binaries.  
  
No VM guests are running outdated hypervisor (qemu) binaries on this host.  
semmy@myserver:~$
```

[2] Install MySQL in Ubuntu Server

1. Untuk memulai instal MySQL bisa dimulai dengan perintah:

➤ ***sudo apt install mysql-server***

2. Selanjutnya aktifkan service MySQL di Ubuntu 22.04 dengan perintah:

➤ ***sudo systemctl start mysql.service***

3. Untuk menambahkan keamanan pada MySQL database, kita bisa menjalankan perintah ***mysql_secure_installation*** yang akan menambahkan beberapa pilihan seperti validasi ***keamanan password***, tidak memberikan akses untuk *remote login* dan membuang sample user.

Disini ada beberapa tahap yang harus kita lakukan, dimulai dari masuk ke MySQL prompt dengan mengetikan perintah berikut:

➤ ***sudo mysql***

➤ **mysql>** ALTER USER 'root'@'localhost' IDENTIFIED WITH mysql_native_password BY '***ganti password anda***';

4. Kemudian keluar dari MySQL prompt

mysql> exit

[2] Install MySQL *in* Ubuntu Server

[4] Setelah metode autentikasi untuk *user root* kita ubah, kemudian kita jalankan perintah berikut untuk keamanan database agar lebih aman:

➤ ***sudo mysql_secure_installation***

NOTE: ADA BEBERAPA PERTANYAAN YANG AKAN DI JAWAB.

[5] Setelah proses *mysql_secure_installation* selesai, selanjutnya kita bisa mengembalikan metoda autentikasi *user root* ke semula yaitu *auth_socket* dengan beberapa perintah seperti berikut:

> ***mysql -u root -p***

..

mysql>ALTER USER 'root'@'localhost' IDENTIFIED WITH auth_socket;

[6] Selanjutnya, kita bisa melihat status MySQL service dengan perintah:

➤ ***systemctl status mysql.service***

Install Latest Version of PHP

Opsi #1: menginstall PHP versi terbaru secara otomatis:

[1] Menginstall PHP sekaligus modul pendukung lainnya:

➤ *sudo apt install php libapache2-mod-php php-mbstring php-xmlrpc php-soap php-gd php-xml php-cli php-zip php-bcmath php-tokenizer php-json php-pear*

[2] Buat “*test.php*” file guna memastikan PHP sudah terinstall di server:

➤ *sudo nano /var/www/html/test.php*

Dan ketikan:

```
<?php  
phpinfo();  
?>
```

[3] Akhiri dengan klik tombol Ctrl + X dan type Y, lalu Enter secara berurutan.

[4] Silakan kembali ke browser dan ketikan “*http://72.168.19.80/test.php*”.
Browser akan memunculkan tampilan PHP version. Berarti kita sudah berhasil.

Install Latest Version of PHP

Opsi #2: jika terjadi error pada cara pertama, gunakan cara yang kedua untuk menginstall PHP versi terbaru secara otomatis:

[1] Menginstall PHP sekaligus modul pendukung lainnya:

➤ *sudo apt install php libapache2-mod-php php-mysql*

[2] Buat “*test.php*” file guna memastikan PHP sudah terinstall di server:

➤ *sudo nano /var/www/html/test.php*

Dan ketikan:

```
<?php  
phpinfo();  
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[4] Silakan kembali ke browser dan ketikan “*http://72.168.19.80/test.php*”. Browser akan memunculkan tampilan PHP version. Berarti kita sudah berhasil.

[3] Create Project Directory

[1] Masuk sebagai administrator atau *root*:

➤ *sudo su*

[2] Buat project folder baru dengan perintah (**P**arent directory):

➤ *sudo mkdir -p /var/www/html/devops*

[3] Konfigurasi permisi hak akses dan pemilik dari folder:

➤ *sudo chmod -R 755 /var/www/html/devops*

➤ *sudo chown -R semmy:semmy /var/www/html/devops*

[4] Restart Apache Web Server:

➤ *sudo systemctl restart apache2*

```
root@unklab: /var/www/html
root@unklab:/home/semmy# sudo mkdir -p /var/www/html/unkpresent
root@unklab:/home/semmy# sudo chmod -R 755 /var/www/html/unkpresent
root@unklab:/home/semmy# cd ..
root@unklab:/home# cd ..
root@unklab:/# ls
bin  dev  home  lib32  libx32  media  opt  root  sbin  srv  tmp  var
boot  etc  lib  lib64  lost+found  mnt  proc  run  snap  sys  usr
root@unklab:/# cd var
root@unklab:/var# ls
backups  crash  local  log  opt  snap  tmp
cache  lib  lock  mail  run  spool  www
root@unklab:/var# cd www/
root@unklab:/var/www# ls
html
root@unklab:/var/www# cd html/
root@unklab:/var/www/html# ls
index.html  test.php  unkpresent
root@unklab:/var/www/html# ls -l
total 20
-rw-r--r-- 1 root root 10671 Mar 26 07:06 index.html
-rw-r--r-- 1 root root 20 Mar 26 07:29 test.php
drwxr-xr-x 2 root root 4096 Mar 26 07:30 unkpresent
root@unklab:/var/www/html# sudo chown -R semmy:semmy /var/www/html/unkpresent
root@unklab:/var/www/html# sudo systemctl restart apache2
root@unklab:/var/www/html# ls -l
total 20
-rw-r--r-- 1 root root 10671 Mar 26 07:06 index.html
-rw-r--r-- 1 root root 20 Mar 26 07:29 test.php
drwxr-xr-x 2 semmy semmy 4096 Mar 26 07:30 unkpresent
root@unklab:/var/www/html#
```

[4] Config File .htaccess in Apache2

Untuk mengaktifkan file *.htaccess* di Apache2 pada Ubuntu Server, ikuti langkah-langkah berikut:

[1] Pastikan modul rewrite sudah diaktifkan. Periksa apakah modul ini sudah diaktifkan dengan menjalankan perintah:

➤ ***sudo a2enmod rewrite***

[2] Setelah modul rewrite diaktifkan, buka file konfigurasi untuk situs web yang ingin Anda aktifkan *.htaccess*. Misalnya, jika Anda ingin mengaktifkan *.htaccess* untuk situs web default, buka file konfigurasinya dengan menjalankan perintah:

➤ ***sudo nano /etc/apache2/sites-available/000-default.conf***

[3] Di dalam blok konfigurasi untuk situs web, tambahkan konfigurasi berikut di bawah direktif **<VirtualHost>**:

<Directory "/var/www/html">

AllowOverride All

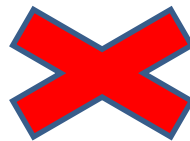
</Directory>

[4] **OR** Jika ingin mengaktifkan *.htaccess* di direktori */home/user/public_html*, maka pada langkah ke-3, gantilah konfigurasi sebagai berikut:

<Directory "/home/semmy/public_html">

AllowOverride All

</Directory>

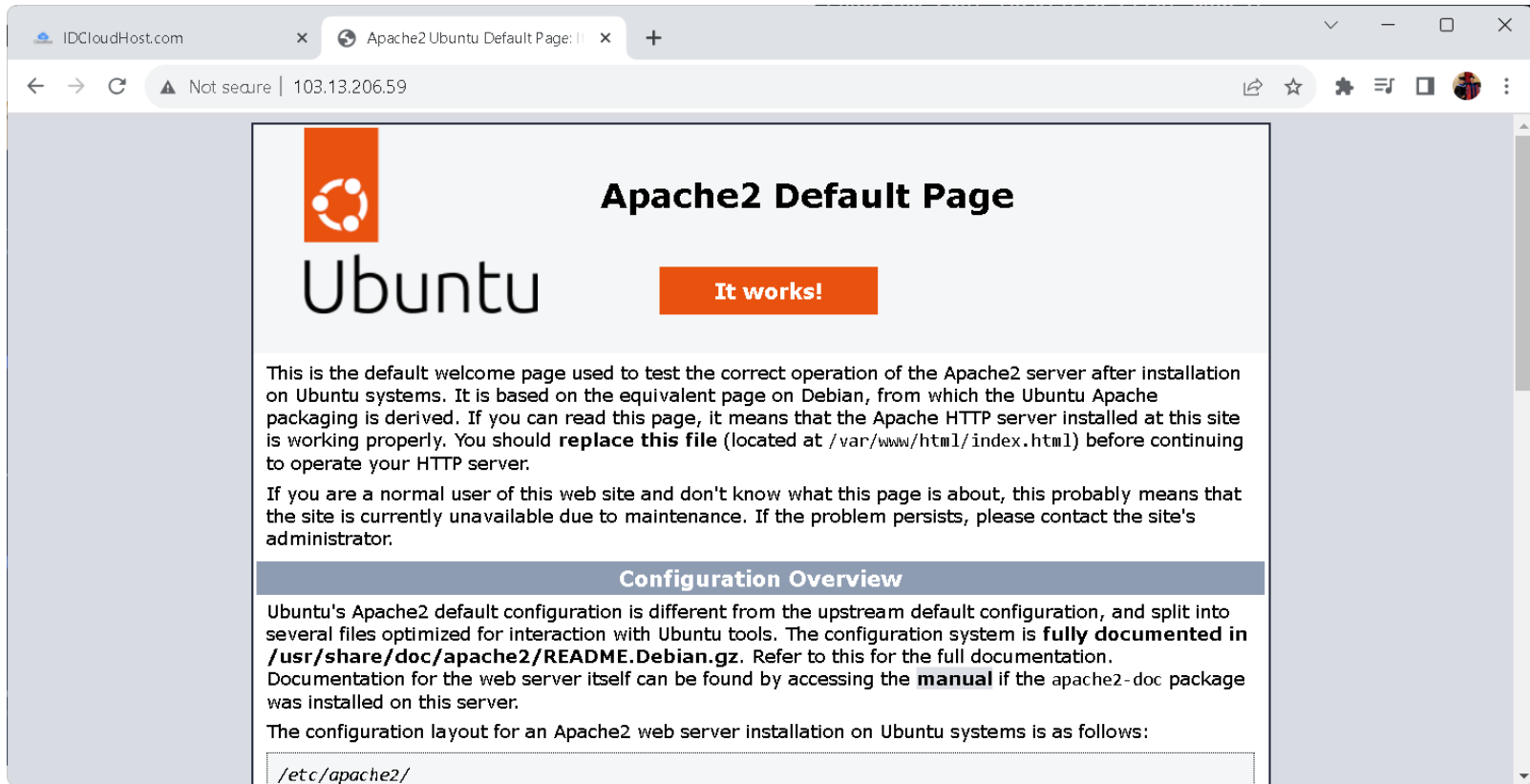


[5] Simpan dan tutup *file* konfigurasi dengan menekan **Ctrl+X**, lalu ketik **Y** untuk menyimpan perubahan dan Enter untuk menutup editor.

[6] Restart apache web server:

➤ ***sudo systemctl restart apache2***

[5] Testing Apache2 Default



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2 – Install & Configure Jenkins

A decorative border of green leaves and branches at the bottom of the page.

[1] Login VPS & Install Dependencies

[1] Change hak akses *superuser* atau *root*:

➤ **sudo su**

[2] Setup SSH untuk generate *private* dan *public* key:

➤ **ssh-keygen**

➤ **Enter passphrase: *******

[3] Install Java versi 11:

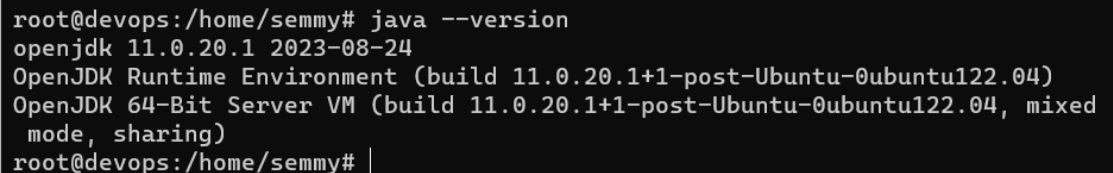
➤ **javac -version (Cek java versi dan rekomendasi)**

➤ **sudo apt-get install openjdk-11-jdk-headless**

➤ Install Git:

➤ **git --version**

➤ **sudo apt-get install git**

A terminal window with a dark background and light text. The title bar shows 'root@devops: /home/semmy' and standard window controls. The terminal content shows the command 'java --version' being executed, resulting in three lines of output: 'openjdk 11.0.20.1 2023-08-24', 'OpenJDK Runtime Environment (build 11.0.20.1+1-post-Ubuntu-0ubuntu122.04)', and 'OpenJDK 64-Bit Server VM (build 11.0.20.1+1-post-Ubuntu-0ubuntu122.04, mixed mode, sharing)'. The prompt 'root@devops:/home/semmy#' is visible at the bottom.

```
root@devops:/home/semmy# java --version
openjdk 11.0.20.1 2023-08-24
OpenJDK Runtime Environment (build 11.0.20.1+1-post-Ubuntu-0ubuntu122.04)
OpenJDK 64-Bit Server VM (build 11.0.20.1+1-post-Ubuntu-0ubuntu122.04, mixed
mode, sharing)
root@devops:/home/semmy#
```

[2] Install & Configure Jenkins

[1] Download Web Application Archive (War) File Jenkins di <https://www.jenkins.io/download/> :

- `sudo wget -c https://get.jenkins.io/war-stable/2.414.1/jenkins.war`
- `sudo curl -O https://get.jenkins.io/war-stable/2.414.1/jenkins.war`

[2] Jalankan Jenkins menggunakan perintah:

- ~~`sudo java -jar jenkins.war --httpPort=9090`~~
- `sudo java -jar jenkins.war --httpPort=9090 & (running background)`

[3] Buka service Jenkins di browser yang sudah running di background:

- `http://103.13.206.59:9090/`

[4] Mendapatkan password secret Jenkins:

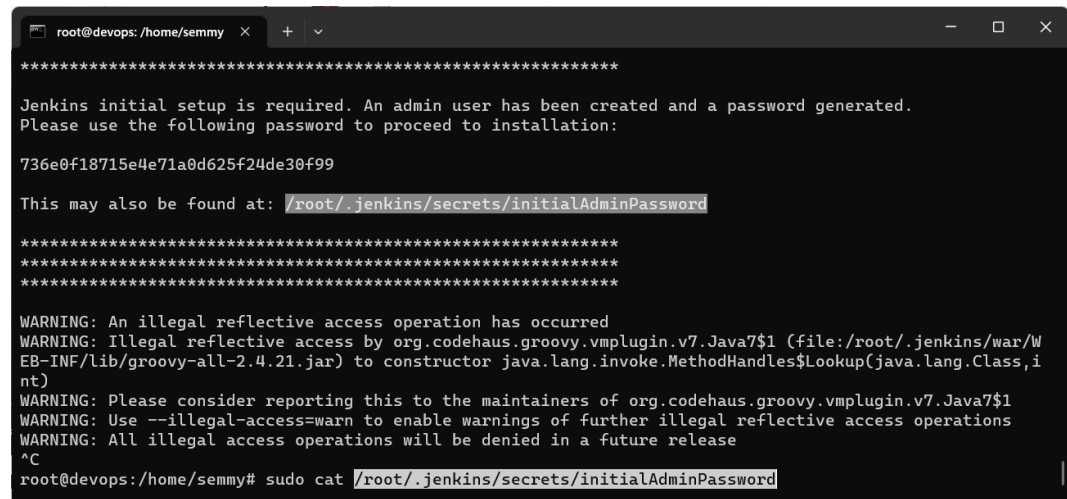
- `sudo cat /root/.jenkins/secrets/initialAdminPassword`

[5] Cek running Jenkins and restart:

- `ps aux | grep jenkins`
- `service jenkins restart`

[6] Firewall setting:

- `sudo ufw enable`
- `sudo ufw allow 22/tcp`
- `sudo ufw allow 80/tcp`
- `sudo ufw allow 443/tcp`
- `sudo ufw allow 9090/tcp`



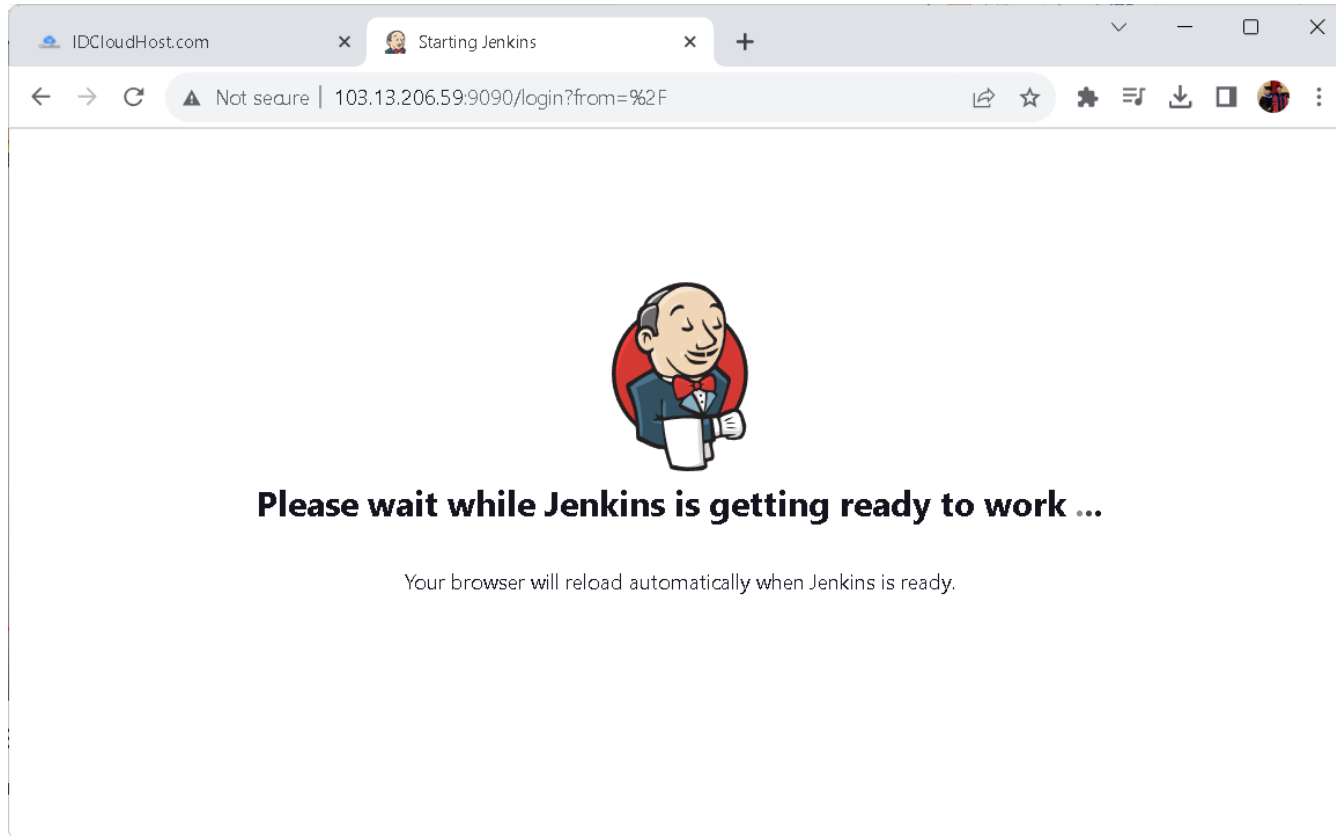
```
root@devops: /home/semmy x + v
*****
Jenkins initial setup is required. An admin user has been created and a password generated.
Please use the following password to proceed to installation:

736e0f18715e4e71a0d625f24de30f99

This may also be found at: /root/.jenkins/secrets/initialAdminPassword

*****
*****
WARNING: An illegal reflective access operation has occurred
WARNING: Illegal reflective access by org.codehaus.groovy.vmplugin.v7.Java7$1 (file:/root/.jenkins/war/WEB-INF/lib/groovy-all-2.4.21.jar) to constructor java.lang.invoke.MethodHandles$Lookup(java.lang.Class,int)
WARNING: Please consider reporting this to the maintainers of org.codehaus.groovy.vmplugin.v7.Java7$1
WARNING: Use --illegal-access=warn to enable warnings of further illegal reflective access operations
WARNING: All illegal access operations will be denied in a future release
^C
root@devops:/home/semmy# sudo cat /root/.jenkins/secrets/initialAdminPassword
```


[3] Waiting *for* Init Jenkins



Ini membutuhkan sedikit waktu untuk proses *init* dari Jenkins !!!
(*kurang lebih 15-20 menit*)

[4] If Server Stop, Restart Jenkins

Saya install di *root directory*, jadi saat restart harus masuk ke directory yang sama. Selanjutnya type:

- **sudo su**
- **sudo java -jar jenkins.war --httpPort=9090 &**

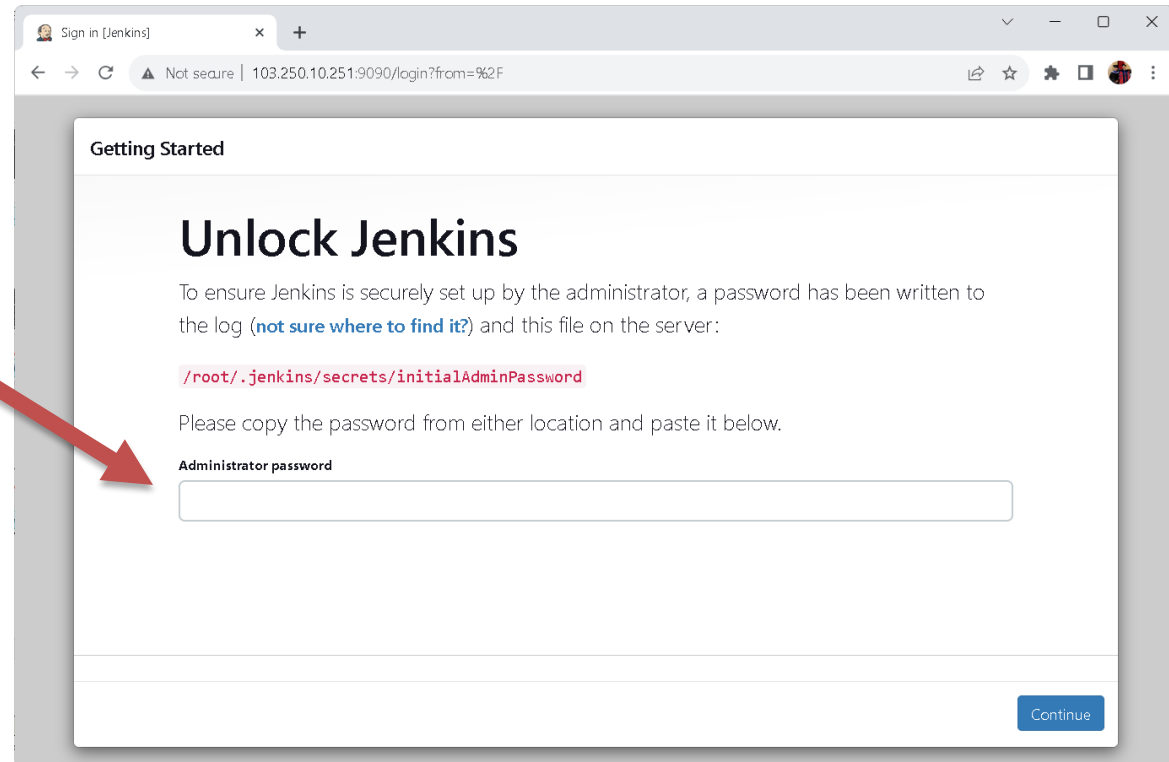
```
root@devops:/home/semmy# ls
root@devops:/home/semmy# ll
total 32
drwxr-x--- 4 semmy semmy 4096 Sep 18 10:04 ./
drwxr-xr-x 3 root  root  4096 Sep 18 09:16 ../
-rw----- 1 semmy semmy    8 Sep 18 10:04 .bash_history
-rw-r--r-- 1 semmy semmy  220 Jan  6  2022 .bash_logout
-rw-r--r-- 1 semmy semmy 3771 Jan  6  2022 .bashrc
drwx----- 2 semmy semmy 4096 Sep 18 09:18 .cache/
-rw-r--r-- 1 semmy semmy  807 Jan  6  2022 .profile
drwx----- 2 semmy semmy 4096 Sep 18 09:16 .ssh/
root@devops:/home/semmy# cd ..
root@devops:/home# cd ..
root@devops:/# ls
bin  etc  lib  libx32  mnt  root  snap  tmp
boot  none  lib32  lost+found  opt  run  srv  usr
dev  jenkins.war  lib64  media  proc  sbin  sys  var
root@devops:/# cd home/
root@devops:/home# cd root
bash: cd: root: No such file or directory
root@devops:/home# cd ..
root@devops:/# sudo java -jar jenkins.war --httpPort=9090 &
```

[4] Unlock Jenkins

Jenkins initial setup is required. An admin user has been created and a password generated. Please use the password to proceed to installation:

➤ **sudo cat /root/.jenkins/secrets/initialAdminPassword**

Copy dan paste
admin password-nya
ke dalam input field
dan click button
continue.



Sign in [Jenkins]

Not secure | 103.250.10.251:9090/login?from=%2F

Getting Started

Unlock Jenkins

To ensure Jenkins is securely set up by the administrator, a password has been written to the log ([not sure where to find it?](#)) and this file on the server:

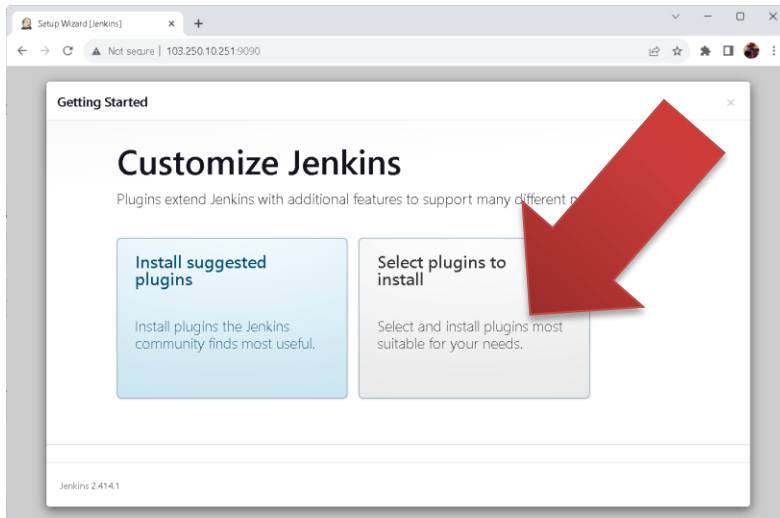
```
/root/.jenkins/secrets/initialAdminPassword
```

Please copy the password from either location and paste it below.

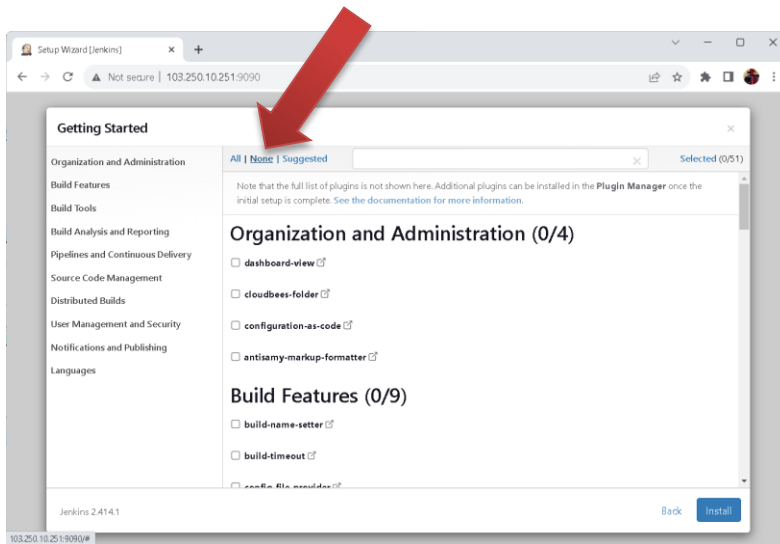
Administrator password

Continue

[5] Customize Jenkins

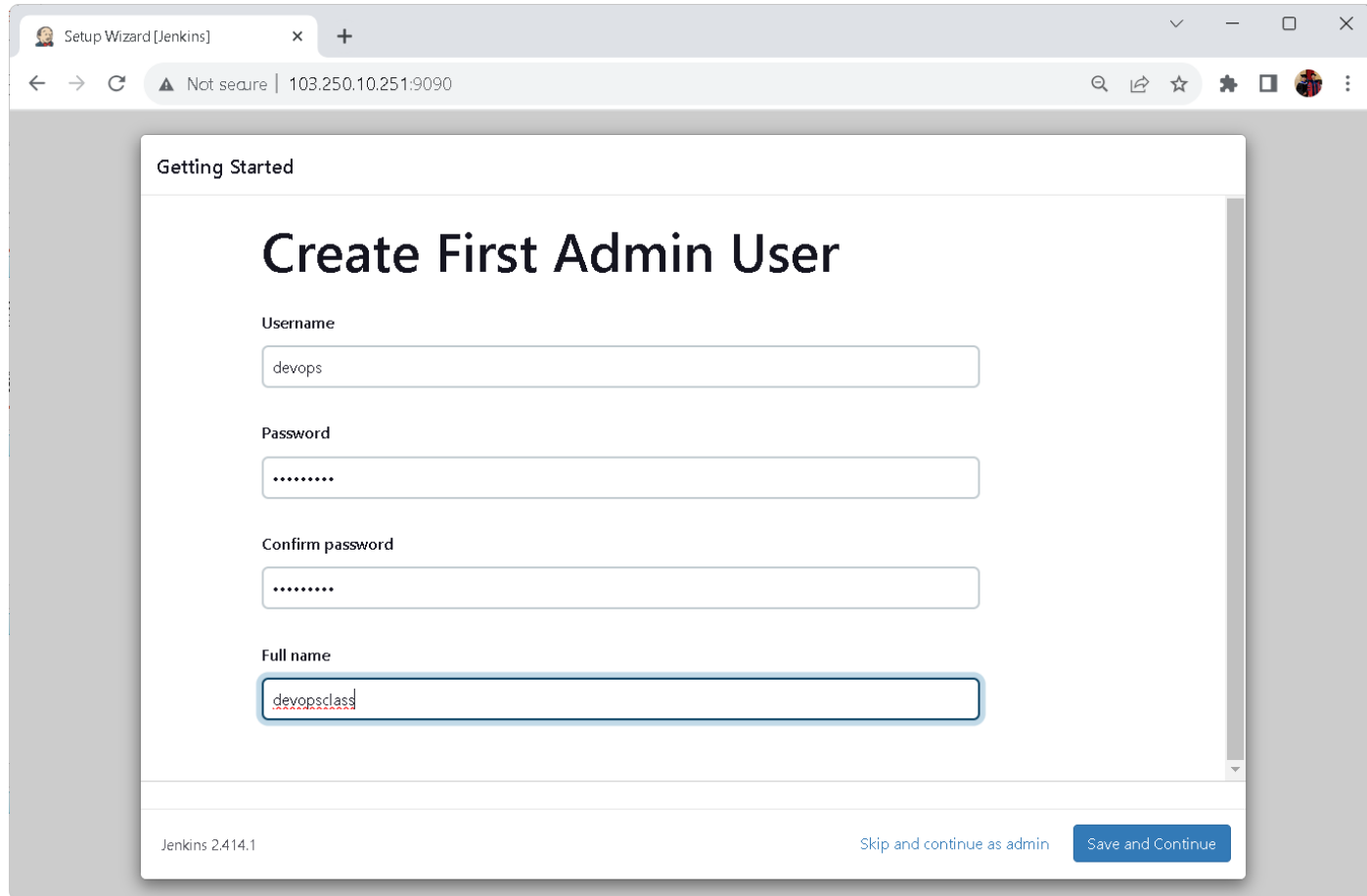


Pilih button “Select plugins to install” untuk bisa memilih opsi yang ada. Tapi pada tahap ini sebenarnya kita tidak ada satupun plugins.



Click pada text link “**None**” yang di tunjukan anak panah merah. Tujuannya supaya tidak satupun plugins yang diinstall pada step ini. Selanjutnya click button Install.

[6] Create Admin User



The screenshot shows a web browser window with the title 'Setup Wizard [Jenkins]'. The address bar shows 'Not secure | 103.250.10.251:9090'. The page content is titled 'Getting Started' and 'Create First Admin User'. It contains four input fields: 'Username' with the value 'devops', 'Password' with masked characters '.....', 'Confirm password' with masked characters '.....', and 'Full name' with the value 'devopsclass'. At the bottom, there is a link 'Skip and continue as admin' and a blue button 'Save and Continue'. The Jenkins version 'Jenkins 2.414.1' is displayed in the bottom left corner.

Setup Wizard [Jenkins]

Not secure | 103.250.10.251:9090

Getting Started

Create First Admin User

Username

Password

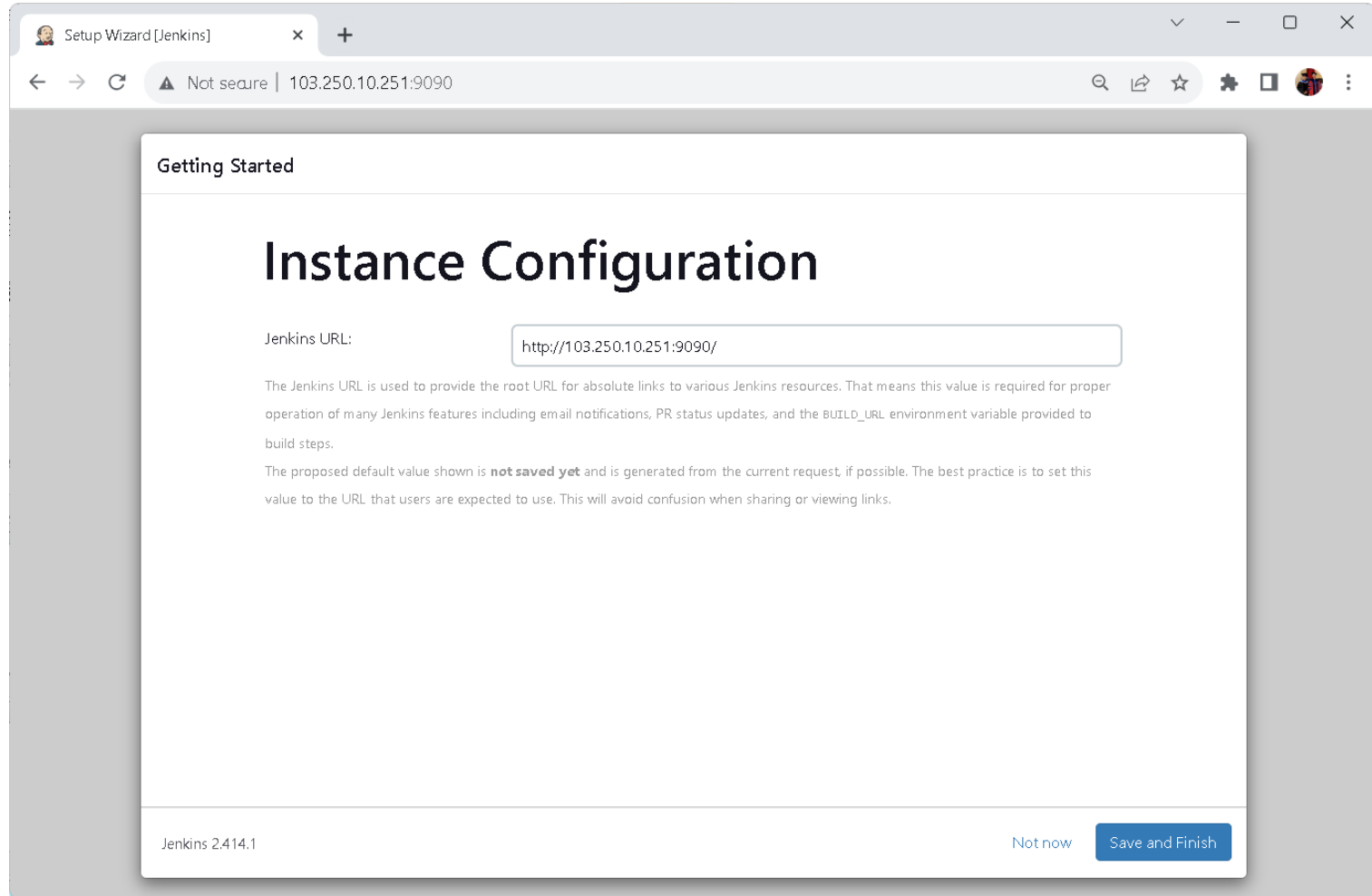
Confirm password

Full name

Jenkins 2.414.1

[Skip and continue as admin](#) [Save and Continue](#)

[7] Instance Configuration



The screenshot shows a web browser window titled "Setup Wizard [Jenkins]". The address bar shows "Not secure | 103.250.10.251:9090". The page content is titled "Getting Started" and "Instance Configuration". It features a text input field for "Jenkins URL:" containing "http://103.250.10.251:9090/". Below the input field, there is explanatory text about the Jenkins URL and its use. At the bottom, there are two buttons: "Not now" and "Save and Finish".

Setup Wizard [Jenkins]

Not secure | 103.250.10.251:9090

Getting Started

Instance Configuration

Jenkins URL:

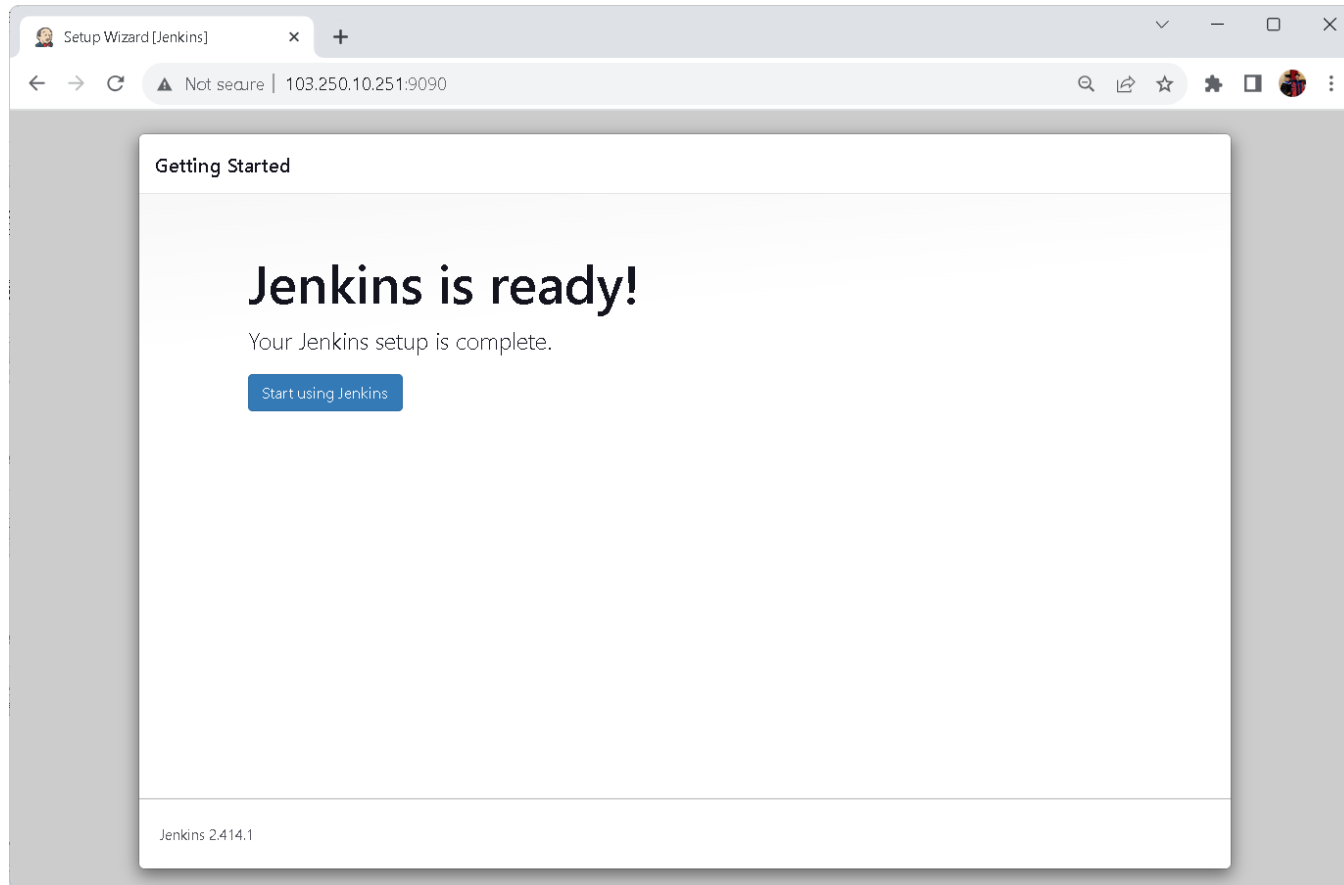
The Jenkins URL is used to provide the root URL for absolute links to various Jenkins resources. That means this value is required for proper operation of many Jenkins features including email notifications, PR status updates, and the `BUILD_URL` environment variable provided to build steps.

The proposed default value shown is **not saved yet** and is generated from the current request, if possible. The best practice is to set this value to the URL that users are expected to use. This will avoid confusion when sharing or viewing links.

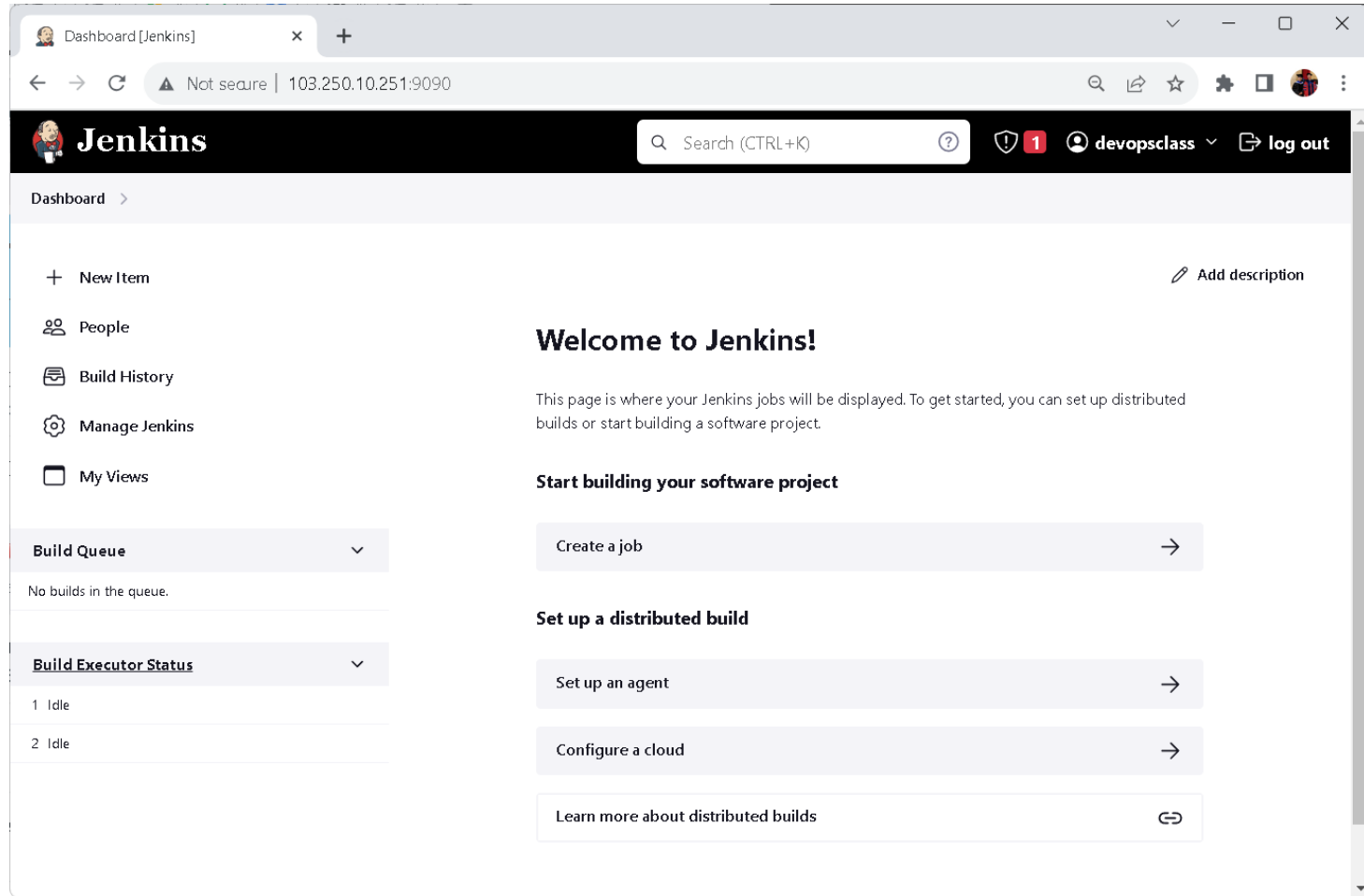
Jenkins 2.414.1

Not now [Save and Finish](#)

[8] Jenkins Setup is Complete



[9] Welcome to Jenkins



The screenshot shows the Jenkins Dashboard in a web browser. The browser's address bar displays "Not secure | 103.250.10.251:9090". The Jenkins header includes the logo, a search bar, a user profile for "devopsclass", and a "log out" button. The left sidebar contains navigation links: "New Item", "People", "Build History", "Manage Jenkins", and "My Views". Below these are two expandable sections: "Build Queue" (showing "No builds in the queue.") and "Build Executor Status" (showing two "Idle" executors). The main content area features a "Welcome to Jenkins!" message, a brief description of the dashboard's purpose, and a "Start building your software project" section with buttons for "Create a job", "Set up a distributed build" (which includes "Set up an agent", "Configure a cloud", and "Learn more about distributed builds").

Dashboard [Jenkins] x +

← → ↺ ⚠ Not secure | 103.250.10.251:9090 🔍 📄 ☆ ⚙️ 📺 👤

Jenkins 🔍 Search (CTRL+K) ? 🛡️ 1 👤 devopsclass ▾ 🚪 log out

Dashboard >

+ New Item ✎ Add description

👤 People

📁 Build History

⚙️ Manage Jenkins

📄 My Views

Build Queue ▾

No builds in the queue.

Build Executor Status ▾

1 Idle

2 Idle

Welcome to Jenkins!

This page is where your Jenkins jobs will be displayed. To get started, you can set up distributed builds or start building a software project.

Start building your software project

Create a job →

Set up a distributed build

Set up an agent →

Configure a cloud →

Learn more about distributed builds ↗

END PRESENTATION

Thank you for your attention

Instructor: S – W – T

THANK YOU

